MILITARY SPECIFICATION

SIGHT, REFLEX, HELICOPTER: XM70

1. SCOPE

1.1 This specification covers one type of reflex sight equipped with a parallelogram linkage to elevate the line of sight in synchronism with a weapon, a height adjustment for the observer, and a range knob for super-elevation angles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10543360	Sight, Reflex: XM70

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PACKAGING DATA SHEETS

10543360	Packaging of Sight, Reflex: XM70
5801065	Packaging of Sight, Reflex, Helicopter: XM70 W/Equipment

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The sight shall be manufactured in accordance with Drawing F10543360.

3.3 General specifications.- Requirements specified herein which are part of each general specification listed below shall be the responsibility of the contractor.

3.3.1 MIL-O-13830:

a. Lenses
b. Cleanliness

3.3.2 MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking
d. Workmanship

3.4 Storage at extreme temperatures.- The sight shall not evidence damage after storage at -65° Fahrenheit (F) and +160°F.

3.5 Performance at -30°F and +150°F.- Requirements 3.5.1 to 3.5.4 inclusive, shall be met while the sight is thermally stabilized at -30°F and at +150°F.

3.5.1 Light intensity control, light switch and height adjustment control.- The light intensity control, light switch, and height adjustment control shall be manually operable.

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3.5.2 Elevation mechanism.- The elevation mechanism shall be operable throughout its excursion range.

3.5.3 Range knob.- The torque required to rotate the knob, through the range specified in 3.6.7 (Table I) shall be 5 to 11 pound-inches.

3.5.4 Arm 10543404.- The arm shall be manually movable from its stowed to operating to stowed location and shall fasten in both the operating and stowed positions.

3.6 Performance between 60°F and 90°F.- Except where otherwise specified, the sight shall meet the requirements of 3.6.1 to 3.6.12 after orientation "a" to "e" inclusive has been effected. (Reference Figure 1 and Drawing F10543360.)

a. The angular difference between line "X" and a perpendicular to line "W" shall be 17° 18' 24".
b. Range knob "RED" line (1/2 inch long) shall be in coincidence with Housing 10543377 index.
c. The engraved line on the main housing shall be in coincidence with the main elevation support index.
d. Arm 10543404 shall be secure in its operating position.
e. Applied electrical potential shall conform to Drawing D10543367.

3.6.1 Vibration "A".- The sight shall show no evidence of damage subsequent to meeting the vibration specified herein, with Bracket 10543391 immobilized, no electrical potential applied, and Arm 10543404 in an operating and stowed position. Vibration shall be as follows:

a. Along three mutually perpendicular planes for a period of three hours in each plane.
b. In accordance with the sinusoidal frequencies and amplitudes in Figure 2.
c. Frequencies in Figure 2 shall vary from 5 to 200 to 5 cycles per second (cps) at a logarithmic rate over a 15 ± 1.0 minute period.
d. A 30 minute dwell shall occur at each (not exceeding four) of the most significant resonant frequencies during the three hour vibration in each plane.

3.6.2 Vibration "B".- The sight shall show no evidence of damage subsequent to meeting the vibration specified herein. There shall be no evidence of sight electrical circuit failure or flickering of the projected reticle image, and there shall be not more than 2.5 mil deviation of the projected reticle image (viewed from the normal operating position) when the sight is vibrated under the following conditions:

a. In a vertical plane with Bracket 10543391 immobilized.
b. At a frequency of 32 cps and an amplitude of 0.015 inch (0.03 inch total excursion) for a period of five minutes ± 15 seconds.

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3.6.3 Azimuth orientation.- The line of sight through the geometric axis of the projected reticle pattern shall be parallel, within ± 4 mils, to a "datum surface" shown on Drawing F10543360, with Arm 10543404 in the operating position.

3.6.4 Elevation orientation.- The line of sight through the geometric axis of the projected reticle pattern shall be parallel to reference line "W" (Figure 1) within 4.0 mils with Arm 10543404 in the operating position and Bracket 10543391 mobilized.

3.6.5 Elevation accuracy.- With the elevation position established in 3.6.4, the angular changes in the line of sight between 213 mils depression and 213 mils elevation shall be within 1.0 mil of the same angular changes in Reference Line "W". The angular changes in the line of sight, between 213 mils depression and 427 mils depression shall be within 2.0 mils of the same angular change in Reference Line "W".

3.6.6 Plumb travel.- The line of sight established in 3.6.4 shall track a plumb line within 1.0 mil from 213 mils elevation to 213 mils depression, and shall track a plumb line within 2.0 mils from 213 mils depression to 427 mils depression.

3.6.7 Superelevation accuracy.- Rotation of the range knob from the elevation position established in 3.6.4 shall cause an angular change in the line of sight within the specified tolerance as indicated in table I.

TABLE I

RANGE (meters)	SIGHT CHANGE (mils)	TOLERANCE (mils)
0 (Red)	0	
250 (White)	-15.38	1.0
500 (White)	- 3.37	1.0
750 (White)	2.86	1.5
1000 (White)	12.70	2.5
250 (Yellow)	-23.93	1.0
500 (Yellow)	19.52	1.0
750 (Yellow)	55.80	1.5
1000 (Yellow)	95.99	1.5
1250 (Yellow)	140.72	1.5
1500 (Yellow)	196.94	2.5

3.6.8 Height adjustment mechanism.- Movement of the height adjustment mechanism throughout its entire excursion shall not cause the line of sight to deviate in excess of 2.0 mils from a plumb line.

3.6.9 Projected reticle pattern definition.- The projected reticle pattern shall be clearly defined with no visible breaks or irregularities, when viewed from the normal operating position in a darkened area or against a bright sky.

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3.6.10 Projected reticle pattern accuracy.- The angle subtended by the circle of the projected reticle pattern shall be 60 ± 2.0 mils.

3.6.11 Parallax.- This requirement shall be met with the elevation position established in 3.6.4. With the observer's eye at a distance of 8 to 10 inches from the center of the beamsplitter, parallax between the reticle image and target on the optical axis, shall not exceed 2.0 mils over the full aperture of the beamsplitter when viewing targets at a range between 500 and 1000 meters.

3.6.12 Electrical controls.- Rotation of the light intensity control knob, from stop to stop in a clockwise (cw) and counterclockwise (ccw) direction, shall gradually vary reticle lamp light intensity in both positions of the filament selector switch.

3.6.13 Sealing.- There shall be no leakage in excess of 0.05 pounds-per-square-inch (psi) from Housing Assembly 10543353, when it is pressurized with an internal pressure of 5 ± 0.2 psi.

3.6.14 Interior atmosphere.- The interior atmosphere of Housing Assembly 10543353 shall have a dew point of -25° Fahrenheit (F).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

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4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II, III and IV herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE II - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE			
MAJOR:	AQL 0.65 percent defective		
101.	Performance between 60°F and 90°F	3.6	4.6.3
102.	Vibration "B"	3.6.2	4.6.3.2
103.	Azimuth orientation	3.6.3	4.6.3.3
104.	Elevation orientation	3.6.4	4.6.3.4
105.	Elevation accuracy	3.6.5	4.6.3.5
106.	Plumb travel	3.6.6	4.6.3.6
107.	Superelevation accuracy	3.6.7	4.6.3.7
108.	Height adjustment mechanism	3.6.8	4.6.3.8
109.	Projected reticle pattern definition	3.6.9	4.6.3.9
110.	Projected reticle pattern accuracy	3.6.10	4.6.3.10
111.	Parallax	3.6.11	4.6.3.11
112.	Electrical controls	3.6.12	4.6.3.12
113.	Sealing	3.6.13	4.6.3.13
114.	Interior atmosphere	3.6.14	4.6.3.14
MINOR:	NONE DEFINED		

NOTE: The tests in Table II shall be conducted at temperatures between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three sights shall be selected at random by a Government representative as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

CHARACTERISTICS		REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.2	Applicable drawings - visual
302.	General specification	3.3.1	MIL-O-13830 - visual
303.	General specification	3.3.2	MIL-F-13926 - visual

4.4.2 Environmental.- One sight shall be selected at random by a Government representative as a special sample from each 25 produced or from

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each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in Table II.

TABLE IV

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
304. Storage at extreme temperatures (-65°F and +160°F)	3.4	4.6.1
305. Performance at -30°F and at +150°F	3.5	4.6.2
306. Light intensity control, light switch, and height adjustment control	3.5.1	4.6.2.1
307. Elevation mechanism	3.5.2	4.6.2.2
308. Range knob	3.5.3	4.6.2.3
309. Arm 10543404	3.5.4	4.6.2.4
310. Vibration "A"	3.6.1	4.6.3.1

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered

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to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.- Test equipment shall be constructed to permit the sight to be positioned as shown on Drawing F10543360. A cross leveling fixture (Drawing D23553) and final inspection fixture (Drawing F10547466) shall be provided to support the sight in the above position, establish specified settings in elevation and provide a means for sight collimation. A wall target or collimator target having acceptance or rejection limitation markings within an accuracy of 0.1 mil shall also be provided. Throughout vibration a means shall be provided to immobilize Bracket 10543391 (see 3.6.1 and 3.6.2).

4.5.2.4 Test target.- A collimator or wall target such as shown in Figure 3 shall be provided. The target shall contain mil blocks (see "E" in Figure 3) to facilitate inspection when determining reticle accuracy or movement, plumb travel, and parallax. When employed as a wall target, the size of a mil block shall be determined as follows:

$$\text{Mil block size} = \frac{\text{Target distance (in inches)}}{1000}$$

The target shall also contain a vertical and horizontal line (cross hair) with a midpoint intersection for zero boresighting (see Line "A" and "B" in Figure 3). The length and thickness of target lines will vary in proportion to target distance. They shall be calculated from the formula:

$$D = \frac{HS}{F}$$

When: D = Distance on target from the geometric axis to any line or mark on target
H = Distance from the geometric axis of reticle to the corresponding line or mark on the reticle
S = Distance from the objective being tested to the target in inches
F = Equivalent objective lens focal length of the reflex sight

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

ITEM	NOMENCLATURE	DESCRIPTION
1.	Torque wrench	Standard shop type readings to be accurate within 10%
2.	Torque adapter	Standard shop type

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ITEM	NOMENCLATURE	DESCRIPTION
3.	Electrical power source	20 to 28 Vdc adjustable output; 8% regulation
4.	Volt-ammeter	0-50 Vdc; 0-1 amp. dc, full scale accuracy ± 2% on both scales

4.6 Test methods and procedures.

4.6.1 Storage at extreme temperatures.- The "Test Facilities" requirements of MIL-F-13926 shall apply except as otherwise specified herein. In lieu of the standard ambient test conditions of MIL-F-13926, the sight shall be thermally stabilized for an uninterrupted period of at least 24 hours at -65°F and at +160°F. Upon completion of this test, the sight shall be examined for evidence of damage.

4.6.2 Performance at -30°F and +150°F.- The "Test Facilities" requirements of MIL-F-13926 shall apply except as otherwise specified herein. In lieu of the standard ambient test conditions of MIL-F-13926, the sight shall be thermally stabilized for an uninterrupted period of at least one hour at -30°F and at +150°F. Tests 4.6.2.1 to 4.6.2.4 inclusive shall be conducted while the sight is thermally stabilized at each temperature specified herein. Upon completion of this test the sight shall be examined for evidence of damage.

4.6.2.1 Light intensity control, light switch and height adjustment control.- Manually rotate the light intensity control knob and height adjustment control knob throughout their respective ranges, in a clockwise (cw) and counterclockwise (ccw) direction. Determine that no binding is evident in either control using normal hand pressure and that the light intensity control knob is secure on its shaft. The light switch toggle when manually operated shall be examined for ease of manipulation with positive indexing in both "ON" positions.

4.6.2.2 Elevation mechanism.- The height adjustment control knob shall be rotated ccw while observing release of the bearing rod and free movement of the elevation mechanism. Manually rotate the height adjustment control knob cw and determine that the elevation mechanism will secure at any desired point throughout its excursion range.

4.6.2.3 Range knob.- To perform this test utilize a standard torque wrench with appropriate adapter. The range knob shall be rotated throughout the ranges in Column 1 of Table I in both a cw and ccw direction. Observe that torque readings over the ranges specified herein, in a cw and ccw direction, remain within the limits specified in 3.5.3.

4.6.2.4 Arm 10543404.- Utilize the force exerted by one hand (with no lever tool) to remove Arm 10543404 from its stowed position and locate it in the operating position. Utilize an equivalent force to

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remove Arm 10543404 from its operating position and return it to a stowed position. Operating and stowed positions shall be as specified on Drawing F10543360. Observe that Arm 10543404 moves freely between operating and stowed positions and determine that the mechanism provided will fasten it in both positions.

4.6.3 Performance between 60°F and 90°F.- The following tests shall be performed in accordance with the "Test Facilities" requirement of MIL-F-13926. Except as otherwise specified in each individual test requirement, test setup and test procedures specified in subparagraphs a and b shall be employed as applicable to meet requirements of 3.6 that necessitate sight positioning along specified optical and geometric lines.

a. Test setup.- The test setup shall utilize a cross-leveling fixture, final inspection fixture and inspection aid (Gunner's Quadrant M1A1) conforming to Drawings D23553, F10547466 and A8658940 respectively. A collimator with cross-hair reticle and a wall target or collimator projector with a suitable reticle target (see Figure 3) will also be required. In addition utilize a regulated electrical power source, standard commercial type direct current (dc) volt-ammeter, 3X (three power) auxiliary telescope and standard time measuring instrument.

b. Test procedure.- This procedure and the final inspection fixture operating instructions (on Drawing F10547466) along with specific instructions in individual test requirements herein, shall be utilized as applicable to perform tests specified herein. Prior to target alignment a bench level may be used to level the horizontal surface plate in a longitudinal and transverse direction. Mount a collimator and coincide its verticle reticle line to a plumb line. With final inspection test fixture mounted, establish coincidence between the plumb line and vertical reticle line, then without disturbing the collimator or coincidence mentioned, raise or lower the target until its horizontal line is coincident with the collimator's horizontal reticle line. Position the sight in test fixture and clamp to avoid shifting. Apply 28 Vdc to the sight and adjust the light intensity control for optimum viewing. Comply with all requirements of 3.6 in accordance with the requirements of 3.6.2 to 3.6.11.

4.6.3.1 Vibration "A".

a. Pretest requirements.- Utilize standard vibration equipment with provision for: Controlling direction of vibration, mounting the sight, and immobilizing Bracket 10543391. A standard time measuring instrument shall be employed to regulate vibration periods. This test shall be performed in compliance with 3.6 (except as otherwise specified in 3.6.1).

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b. Test procedure.- During the first hour of each three hour vibration in each direction, Arm 10543404 shall be in a stowed position for one half hour and in an operating position for the remaining half hour. Record throughout the first hour of vibration in each direction all predominant resonant frequencies exhibited by the sight and note the position of Arm 10543404 at that time. Upon completion of the first hour of vibration in each direction, select the four most prominent resonant frequencies of all those recorded regardless of the position of Arm 10543404. During the remaining 2 hour vibration in each direction, vibrate the sight for not less than 30 minutes at each one of the four resonant frequencies previously selected with Arm 10543404 in the position noted when the resonant frequency was first recorded. Upon completion of the entire vibration test (of three hours in each direction for a total of nine hours), the sight shall be given an examination for evidence of physical damage.

4.6.3.2 Vibration "B".- Utilize standard vibration equipment with provision for mounting sight in compliance with 3.6, immobilizing Bracket 10543391 and vibrating the sight in a vertical plane only. A standard time measuring instrument (to regulate vibration periods) and standard commercial type direct current (dc) volt ammeter (to monitor input voltage and current variations) shall be employed. This test shall also utilize an electrical power source and target (both off-table mounted). The target shall have a cross hair reticle whose intersecting point shall be the center of a 5.0 mil diameter circle (see Figure 3).

a. Pretest set-up.- Prior to conducting the actual test apply 28 Vdc in proper polarity (see Drawing D10543367) to the sight making necessary electrical connections to permit input voltage and current readings to be monitored. The observer shall locate his eye eight to ten inches from the beamsplitter in the direction of the sight housing, along the line of sight depicted on Drawing F10543360. The light intensity control knob shall be adjusted for optimum definition of the projected reticle image. Coincidence shall now be established between the geometric center of the projected reticle image and the geometric center of the cross hair reticle on the target prior to vibration.

b. Test procedure.- Vibrate the sight in compliance with 3.6.2, and observe throughout the entire vibration period no displacement in excess of 2.5 mils in the coincidence previously established. Position the filament selector switch of the sight in one "ON" position during half the vibration period and in the alternate "ON" position during the remaining half. Observe with the switch in both positions that there is no input current change indicative of intermittent electrical circuit continuity, electrical short circuit, or reticle lamp failure. Upon completion of this test the sight shall be given a visual and tactile examination for physical damage.

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4.6.3.3 Azimuth orientation.- Utilize a target having a series of mil blocks, each one mil square, distributed evenly each side of a vertical plane on which rests the sight's geometric line of sight. This plane shall be parallel to the datum surface and 90° to a plane on which rests line "X" in Figure 1. The angular distance specified herein shall be measured on a plane parallel to the plane on which rests reference line "W" in Figure 1. Observe that the line of sight through the geometric axis of the projected reticle pattern is within the tolerance specified in 3.6.3.

4.6.3.4 Elevation orientation.- Utilize the inspection aid (see 4.6.3) and a target having a series of mil blocks, each one mil square, distributed evenly each side of a plane parallel to the plane on which rests reference line "W" in Figure 1. Observe that the line of sight through the geometric axis of the projected reticle pattern is within the tolerance specified in 3.6.4.

4.6.3.5 Elevation accuracy.- This test shall be performed subsequent to conducting the azimuth and elevation orientation tests and after compliance with requirements 3.6.3 and 3.6.4. Utilize a target having the required number of mil blocks, each one mil square, distributed evenly each side of the horizontal target line. In accordance with final inspection fixture operating instructions, secure the sight at the elevation station and at the stations in depression specified. Observe that the line of sight through the geometric axis of the projected reticle image is in coincidence with its respective horizontal target line, within the tolerance specified in 3.6.5.

4.6.3.6 Plumb travel.- This test shall be performed subsequent to conducting the azimuth and elevation orientation tests and after compliance with requirements 3.6.3 and 3.6.4. Utilize a target having the required number of mil blocks, each one mil square, distributed evenly each side of a plumb target line. In accordance with final inspection fixture operating instructions release and permit the elevation mechanism to travel the excursion in elevation and depression specified. Observe that the line of sight through the geometric axis of the projected reticle image is in coincidence with the plumb target line within the tolerance specified in 3.6.6.

4.6.3.7 Superelevation accuracy.- Subsequent to test 4.6.3.4 and compliance with 3.6.4, utilize the range knob position established therein and the appropriate station located along a plumb target line (see Figure 3). Rotate the range knob from the above specified position to each range position specified in Table I. To determine compliance with 3.6.7, observe that the resultant angular change in the line of sight is within the tolerance indicated for each induced sight change in accordance with Table I. Observe that each detent exhibits positive indexing. Observe compatibility between range employed and knob scale engraved numerals and other markings.

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4.6.3.8 Height adjustment mechanism.- Subsequent to test 4.6.3.4 and compliance with 3.6.4, conduct the following test. Utilize a target and adjust the sight's line of sight to eye level height for various observers throughout a height adjustment range that will encompass the maximum travel possible for the sight. To determine compliance with 3.6.8 observe in a mirror, provided with the test fixture, no deviation in the line of sight greater than the amount specified when the sight is periodically clamped and sighted throughout the entire excursion.

4.6.3.9 Projected reticle pattern definition.- Rotate the light intensity control knob to obtain optimum resolution of the projected reticle pattern image in a darkened room (with not more than 0.01 foot candle of illumination). The observer shall view the projected reticle pattern image through a 3X auxiliary telescope and determine that it is clearly defined with no breaks or irregularities. After a lapse of at least 15 minutes (light adaptation period) again rotate the light intensity control knob to obtain optimum resolution of the projected reticle pattern image against a bright sky (a uniformly illuminated field of view having a brightness of at least 2300 foot lamberts). Employ the procedure previously specified in this test and make the identical observation to determine compliance with 3.6.9.

4.6.3.10 Projected reticle pattern accuracy.- Rotate the light intensity control knob to obtain optimum resolution of the projected reticle pattern image. Utilize a target circle having the required number of mil blocks, each one mil square, distributed evenly each side of the circle and in each of its quadrants. The diameter of the target circle shall be the chordlength of an arc which intersects both legs of a 60 mil angle. When using a distant target, actual target measurements shall be computed in accordance with 4.5.2.4. The geometric center of the target circle shall be made coincident with the geometric center of the projected reticle pattern image at the beamsplitter. Observe coincidence in all four quadrants between the circle of the projected reticle pattern image and the target within the tolerance specified in 3.6.10.

4.6.3.11 Parallax.- The observer shall locate his eye eight to ten inches from the beamsplitter in the direction of the sight housing along the line of sight depicted by Drawing F10543360. Adjust light intensity control knob for optimum viewing. Align both the vertical and horizontal reticle line of the sight close to but not touching the vertical and horizontal lines on targets, located at an actual or simulated distance of both 500 and 1000 meters. The observer shall slightly move his head up and down and from right to left, while observing both target lines and both reticle lines. The observer shall discover, by target mil block examination, no line displacement in excess of the limit specified over the full aperture of the beamsplitter when viewing the 500 meter and 1000 meter target.

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4.6.3.12 Electrical controls.- Utilize a standard commercial type direct current ammeter to monitor and measure the electrical current drawn by the sight during this test. To determine compliance with 3.6.12, position filament selector switch in one "ON" position and rotate light intensity control knob cw from stop to stop and observe that the reticle lamp will progressively increase the intensity of emitted light. Position filament selector switch in the alternate "ON" position and observe a fluctuation in input current as the switch is actuated. Rotate light intensity control knob ccw and observe that the emitted light progressively attenuates to minimum. Observe compatibility between knob performance and markings and that knob is secure to its shaft. Observe positive switch indexing and ease of manipulation of both controls.

4.6.3.13 Sealing.- Pressurize Housing Assembly 10543353 (Drawing D10543353) with dry nitrogen gas to the value specified in requirement 3.6.13. Retain applied internal pressure for at least one hour, and observe that a pressure gage indicates no pressure drop, in excess of 0.05 pounds-per-square-inch, to assure compliance with 3.6.13.

4.6.3.14 Interior atmosphere.- Subsequent to test 4.6.3.13 and compliance with 3.6.13, bleed Housing Assembly 10543353 to attain an internal pressure of one pound-per-square-inch gage (psig). Flush housing with dry nitrogen gas through a regulator that will prevent housing internal pressure from exceeding one psig during the flushing operation. Continue the flushing operation until it is observed that the nitrogen gas bled from the housing exhibits a dew point at least as low as the value specified in 3.6.14, as measured with a dew point tester. Upon completion of this test the sight shall be sealed to assure compliance with 3.6.14.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking of Reflex Sight: XM70 without equipment (see 5.2) shall be in accordance with Packaging Data Sheet 10543360. The level of protection shall be as specified in the procurement document.

5.2 Alternate packaging, packing and marking.- Packaging, packing and marking of Reflex Sight: XM70 with equipment shall be in accordance with Packaging Data Sheet 5801065. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The reflex sight covered by this specification is primarily intended for use by the pilot of a light observation helicopter (L.O.H.) to obtain a high probability first round hit. It is intended to be installed on the aircraft's centerline (℄) and be energized from the aircraft's internal electrical system. A mechanical linkage connected

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to the gun drive moves the reflex sight in synchronism with the weapon axis in elevation and depression. The reflex sight is fixed in azimuth.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Order of document precedence.
c. Applicable Packaging Data Sheet (see 5.1 and 5.2).
d. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
e. Applicable stock number.

Custodian:

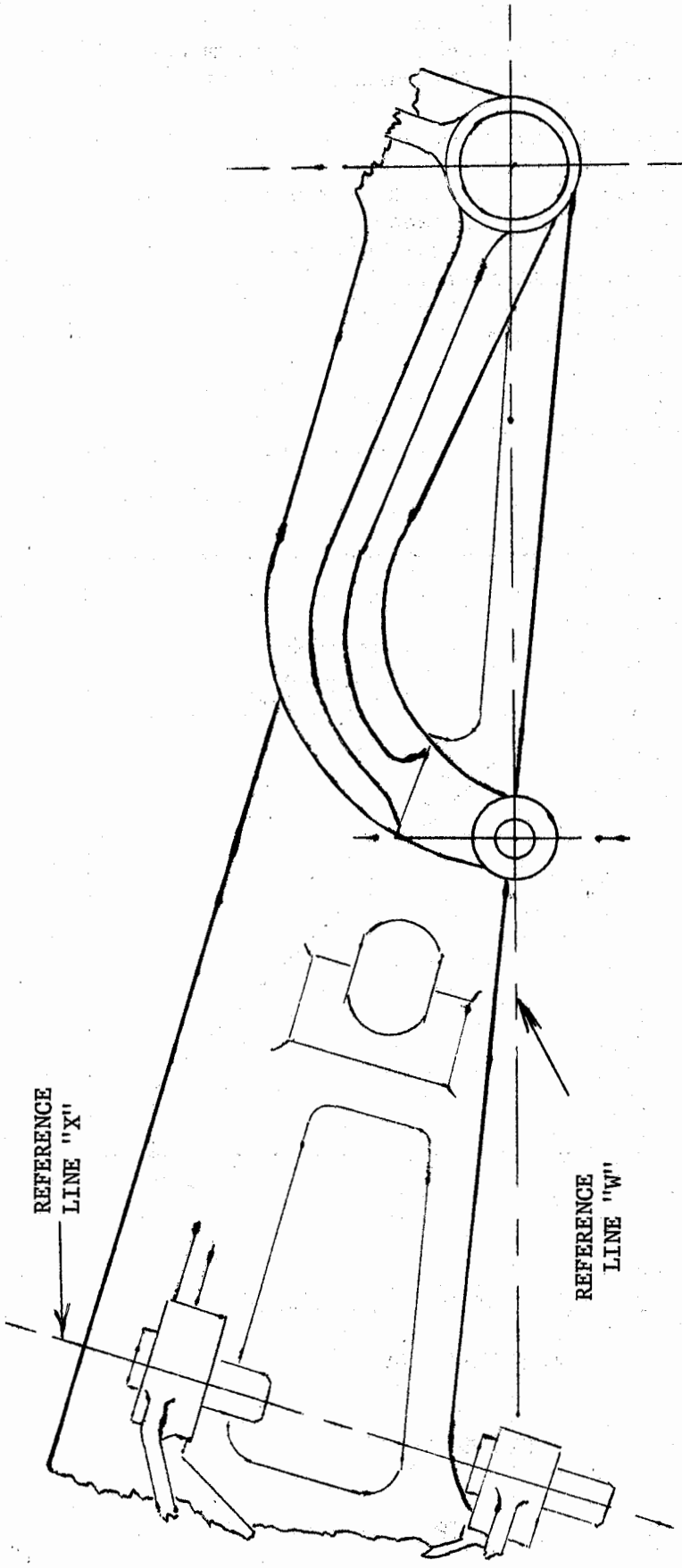
Army - MU

Preparing activity:

Army - MU

Project No. 1270-A041Y

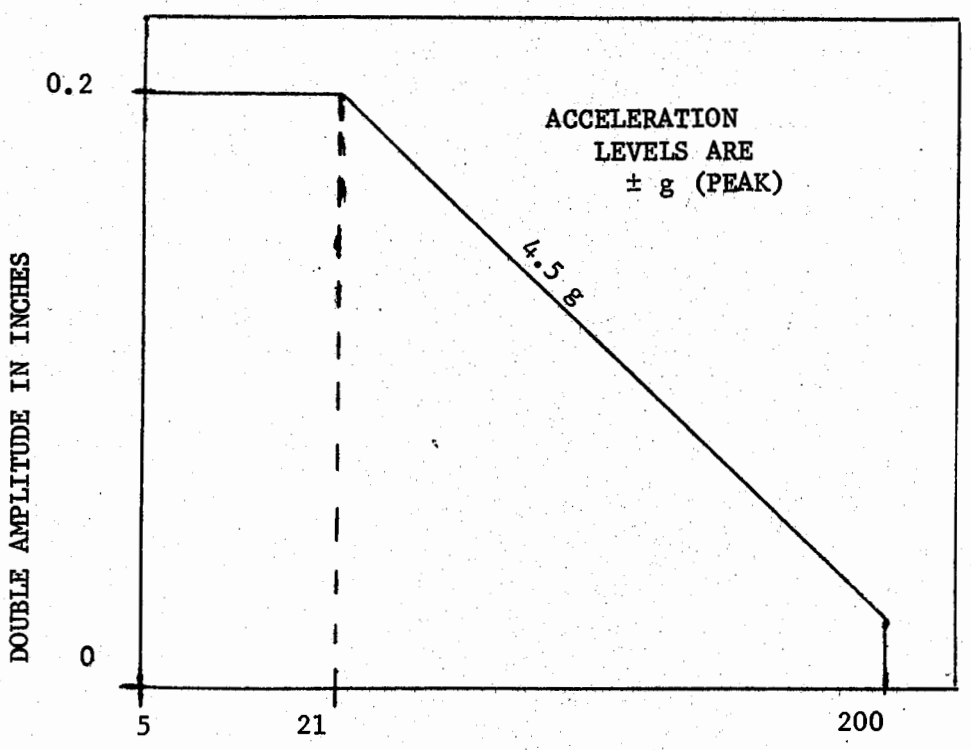
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SUPPORT AND BRACKET ASSEMBLY

FIGURE 1

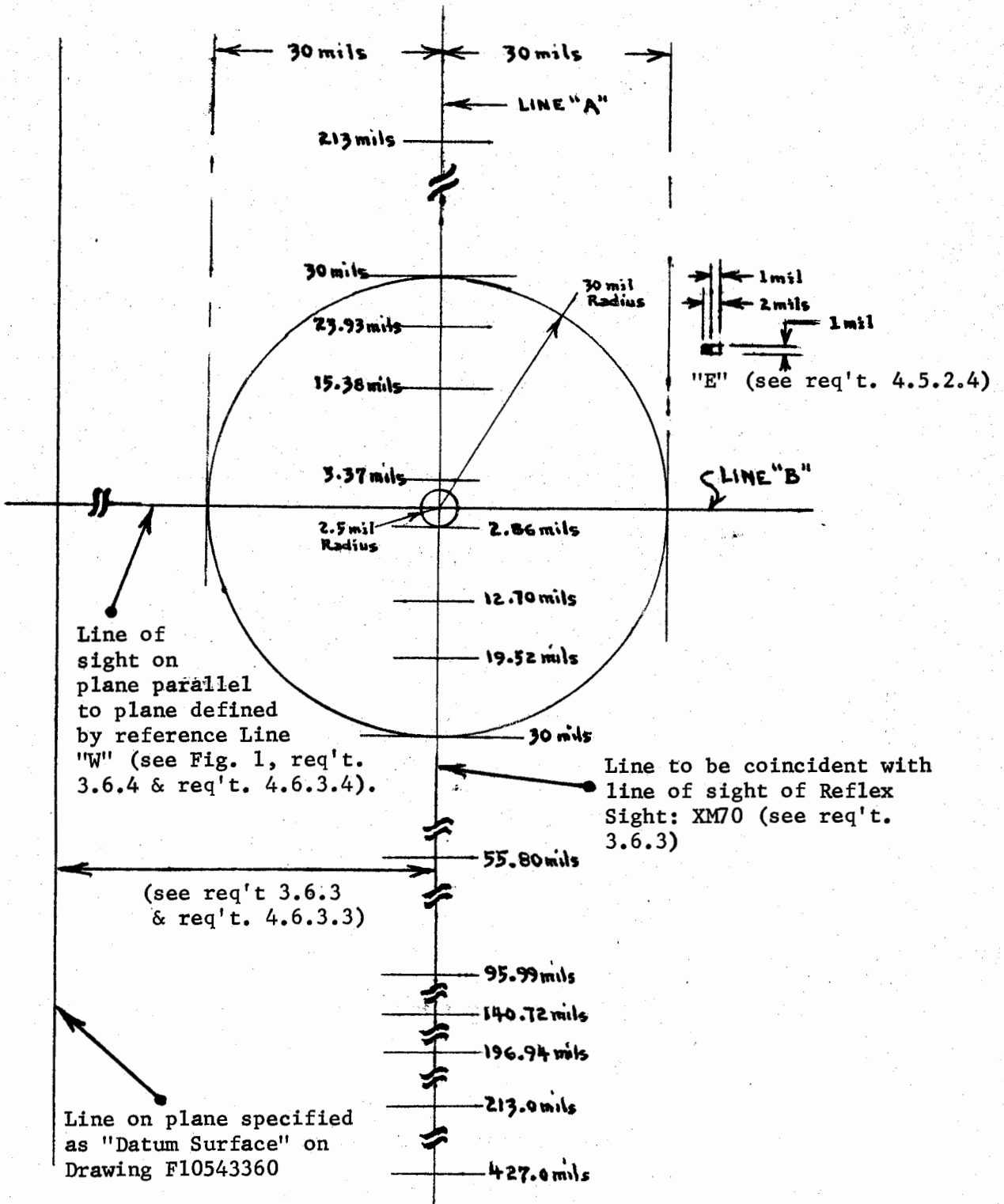
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VIBRATION "A"

FIGURE 2

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NOTES:
1. Target stations to be made equal to chord lengths of arcs which intersect legs of angles having angular values specified.
2. Actual target measurements to be computed in accordance with requirements of 4.5.2.4.
3. Intersection of target lines "A" & "B" to be made coincident with intersection of cross hair on imaged reticle at beam-splitter in target alignment.

FIGURE 3 - TEST TARGET

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